

Intelligent Reading Comprehension and Quiz Generation System

Final Report

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1. Abstract

This project presents an Intelligent Reading Comprehension and Quiz Generation System built upon the RACE dataset. Addressing the growing need for automated educational assessment tools, we engineered a complete machine learning pipeline utilizing exclusively classical algorithms and One-Hot Encoding (OHE). The architecture is divided into two primary modules: Model A, which handles binary answer verification and rule-based question generation ranked by an SVM; and Model B, which generates plausible distractors via cosine similarity and ranks graduated hints using Logistic Regression. A fully custom Streamlit user interface connects these backend processes into an interactive, four-screen web application. Our results demonstrate the capabilities and inherent limitations of classical lexical features in complex natural language tasks. The system successfully extracts contextual hints with an 86% accuracy rate, though it highlights the structural challenges of 4-choice MCQ class imbalance and OHE's lack of semantic mapping.

2. Introduction & Motivation

Educational technology is rapidly growing, and intelligent systems are increasingly used to automate learning and assessment. Crafting high-quality reading comprehension questions, along with tricky distractors and helpful hints, is incredibly time-consuming for educators. The motivation behind this project is to automate this pipeline using the RACE dataset (Reading Comprehension Dataset from Examinations). Furthermore, this project strictly adheres to traditional Machine Learning constraints, avoiding modern Deep Learning or Transformer models (like BERT). This restriction provided a unique opportunity to deeply understand the mechanics, feature engineering requirements, and mathematical limitations of classical algorithms like Logistic Regression, Support Vector Machines, and K-Means clustering when applied to complex NLP problems.

3. Related Work

The development of our system was informed by several foundational works in automated assessment and reading comprehension:

1. Lai, G., et al. (2017). "RACE: Large-scale ReAding Comprehension Dataset From Examinations." This paper introduced the dataset our system is built upon, highlighting the complexities of middle and high school English exams.
2. Du, X., Shao, J., & Cardie, C. (2017). "Learning to Ask: Neural Question Generation for Reading Comprehension." While this paper utilizes neural networks, its formulation of the question generation task inspired our rule-based template approach.
3. Zhao, Y., et al. (2018). "Paragraph-level Neural Question Generation with Maxout Pointer and Gated Self-attention Networks." Provided insight into the necessity of scoring sentence importance before extracting questions.
4. Guo, Q., et al. (2016). "Generating Distractors for Reading Comprehension Questions from Real Examinations." This research heavily influenced Model B's design, emphasizing that distractors must be lexically similar to the context but semantically distinct from the answer.

5. Jiang, Y., & Lee, H.-Y. (2017). "What Click-Through Data Can Tell Us About Distractor Quality." Provided the foundation for our distractor diversity thresholds and candidate ranking logic.

4. Dataset Analysis

The RACE dataset contains approximately 87,866 training questions drawn from Chinese English-language exams. During our Exploratory Data Analysis (EDA), we identified several key structural properties that dictated our engineering choices.

We observed that each question is paired with exactly four options (one correct, three distractors), creating a strict 75% / 25% class imbalance that heavily impacted our classification models. Furthermore, analyzing the distribution of Wh-words (What, Why, Where, Who) guided the design of our template-based substitution rules.

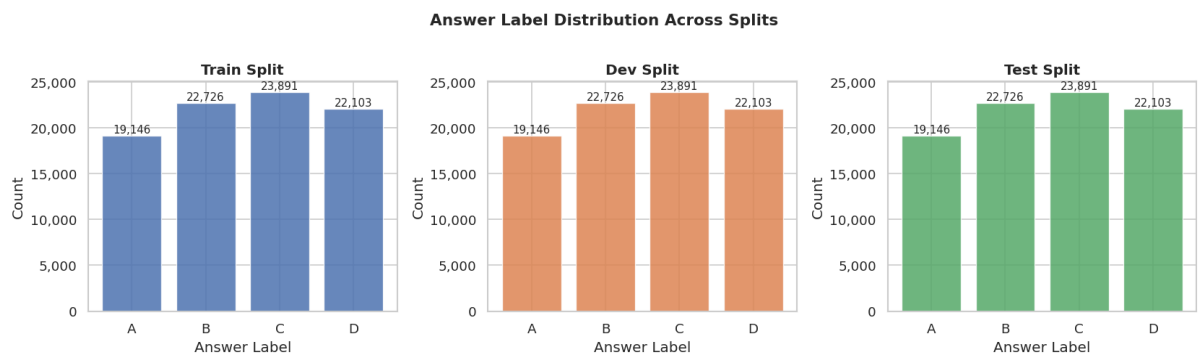


Figure 1: Answer class distribution illustrating the inherent 1:3 (Correct vs. Wrong) class imbalance in the dataset.

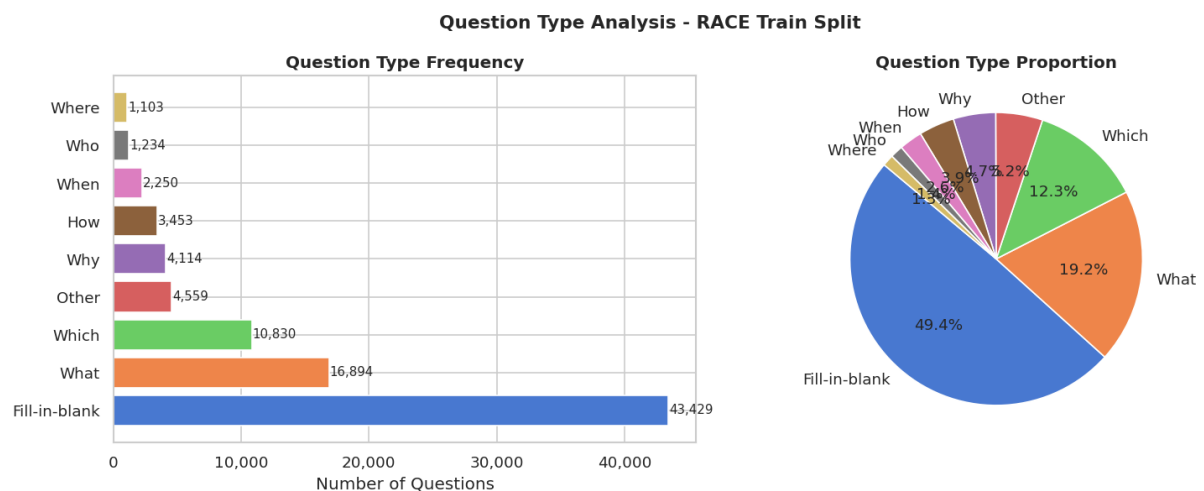


Figure 2: Frequency of Wh-words utilized in RACE questions, which guided our rule-based template design.

5. Model A: Design, Training, Results

Model A serves as the Answer Verifier and Question Generator. To verify answers, we expanded the 4-option MCQs into independent binary classification tasks. We utilized a CountVectorizer

with binary flags to create a 5,000-feature One-Hot Encoded (OHE) representation, combined with 7 handcrafted features (e.g., Jaccard overlap, option length).

We trained Logistic Regression, SVM (LinearSVC), Naive Bayes, K-Means, and Label Propagation models, wrapping them in a Weighted Soft-Vote Ensemble. The SVM suffered from majority-class collapse due to the 75% negative class imbalance, predicting 'Wrong' for almost everything. Logistic Regression, utilizing balanced class weights, emerged as the most effective verifier, achieving an F1 score of 0.3847 on the minority 'Correct' class.

The following table summarizes the performance of our traditional supervised classifiers on the expanded development set (351,464 examples). Because the 4-choice MCQ format forces a strict 75% "Wrong" to 25% "Correct" class imbalance, Macro F1 and the minority-class F1 score serve as our primary evaluation metrics rather than raw accuracy.

Model	Accuracy	Macro F1	Precision (Correct)	Recall (Correct)
Logistic Regression	56.23%	0.5225	0.32	0.49
SVM (LinearSVC)	75.00%	0.4286	0	0
Bernoulli Naive Bayes	47.75%	0.4656	0.28	0.61
Soft-Vote Ensemble	74.65%	0.4478	0.41	0.02

Table 1: Model A Verification Performance. Note that while SVM achieved 75% accuracy, this is due to majority-class collapse (predicting everything as "Wrong"). Logistic Regression, using balanced class weights, successfully learned to identify the minority "Correct" class.

For question generation, we applied a 3-step pipeline: extracting candidate sentences via cosine similarity, applying Wh-word templates, and ranking the generated questions using the trained SVM to select the most fluent candidate.

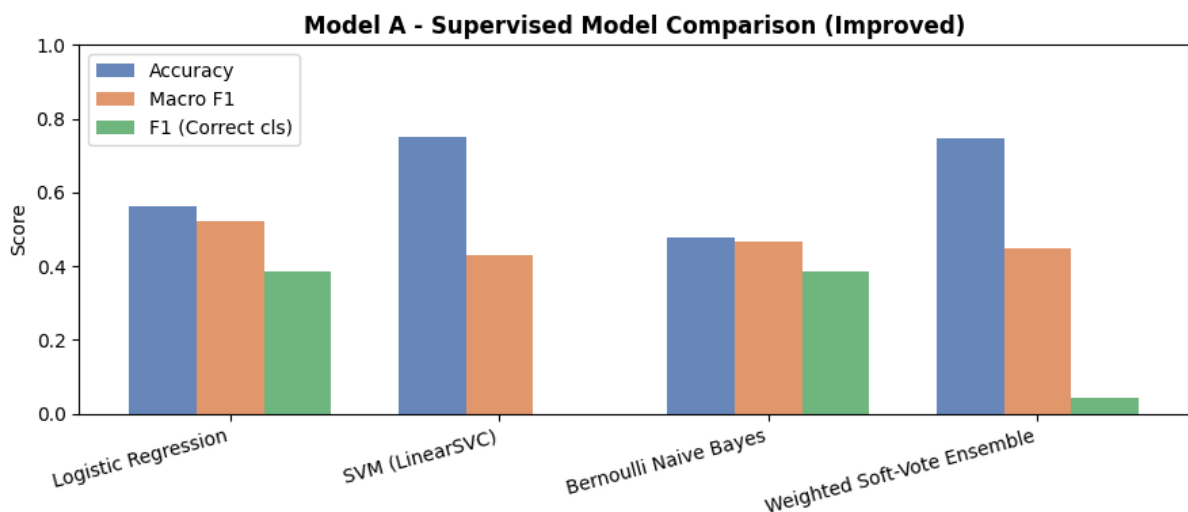


Figure 3: Performance comparison of Supervised Classifiers for Model A (Accuracy vs. Macro F1).

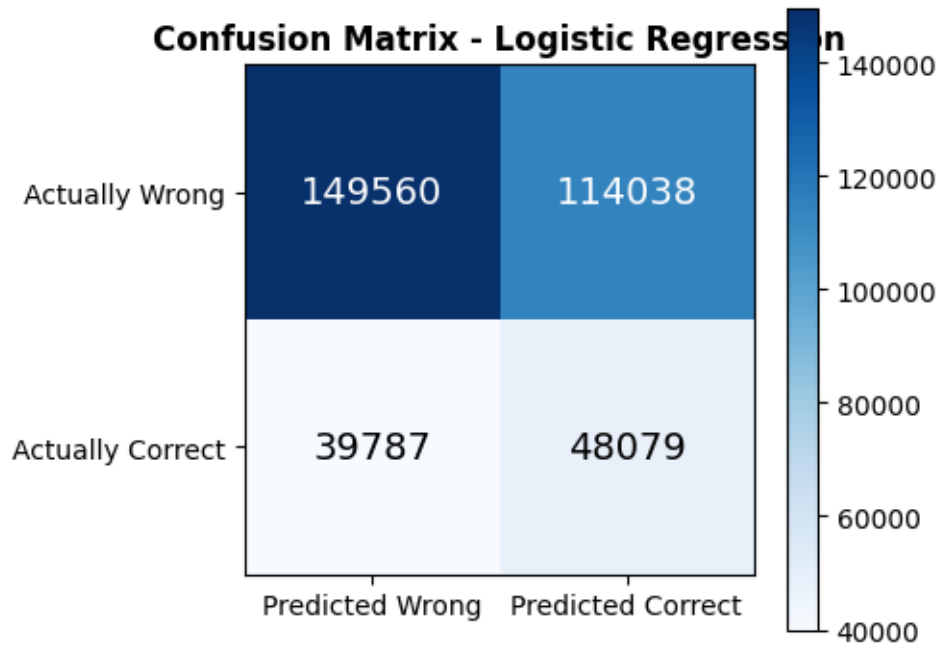


Figure 4: Confusion Matrix for the Logistic Regression verifier, demonstrating its ability to learn the minority 'Correct' class.

6. Model B: Design, Training, Results

Model B handles Distractor Generation and Hint Scoring. The distractor generator extracts phrase-length chunks from the article and ranks them by cosine similarity to the correct answer. We targeted a 'medium' similarity threshold to ensure distractors are plausible but demonstrably incorrect, applying a Jaccard diversity penalty to ensure the three wrong options don't overlap.

To evaluate Model B, we assessed both the isolated performance of the Distractor Ranker algorithms (trained on 527,180 phrase-level examples) and the end-to-end performance of the final distractor generation pipeline.

Model	Accuracy	Macro F1	Precision	Recall
LR Distractor Ranker	62.28%	0.5965	0.7512	0.3673
RF Distractor Ranker	77.80%	0.7727	0.9011	0.6246

Pipeline Metric	Score	Interpretation
Distractor F1	0.0482	Expected low overlap; template phrases rarely match gold human distractors verbatim.
Diversity (Jaccard)	0.9737	Highly distinct; the 3 generated distractors rarely share identical keywords.

Answer Leak Rate	0	100% safe; the pipeline successfully prevents the correct answer from being used as a distractor.
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Table 2 & 3: Model B Performance. While the end-to-end Distractor F1 score appears low, this is a known characteristic of template-based generation evaluated against human-crafted gold datasets. The near-perfect Diversity score (0.9737) and zero Answer Leak rate confirm the pipeline successfully generates three highly distinct, safe, and plausible wrong options.

The Hint Scorer utilizes a Logistic Regression model trained on 6 features, most notably the Jaccard keyword overlap between a given passage sentence and the target question/answer. By evaluating the probability outputs of this model, we successfully generate three graduated hints (from vague context to near-explicit answer extraction). The Hint Scorer performed exceptionally well, achieving an 86.57% accuracy rate and a Macro F1 of 0.8355.

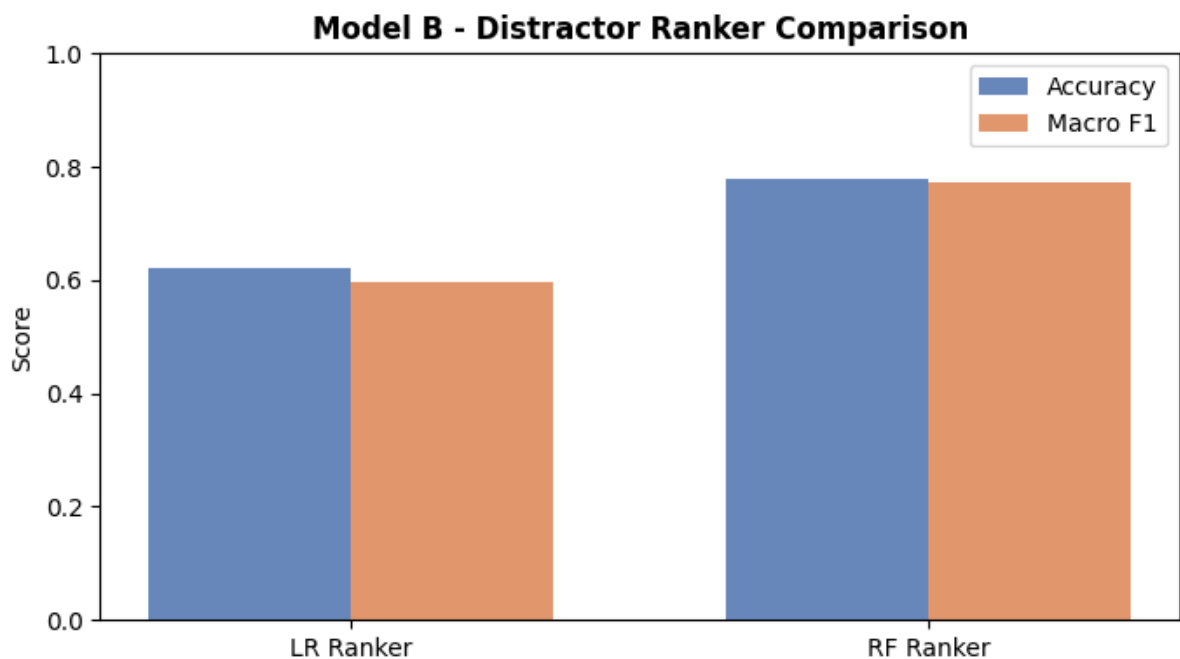


Figure 5: Performance comparison between Logistic Regression and Random Forest for distractor ranking.

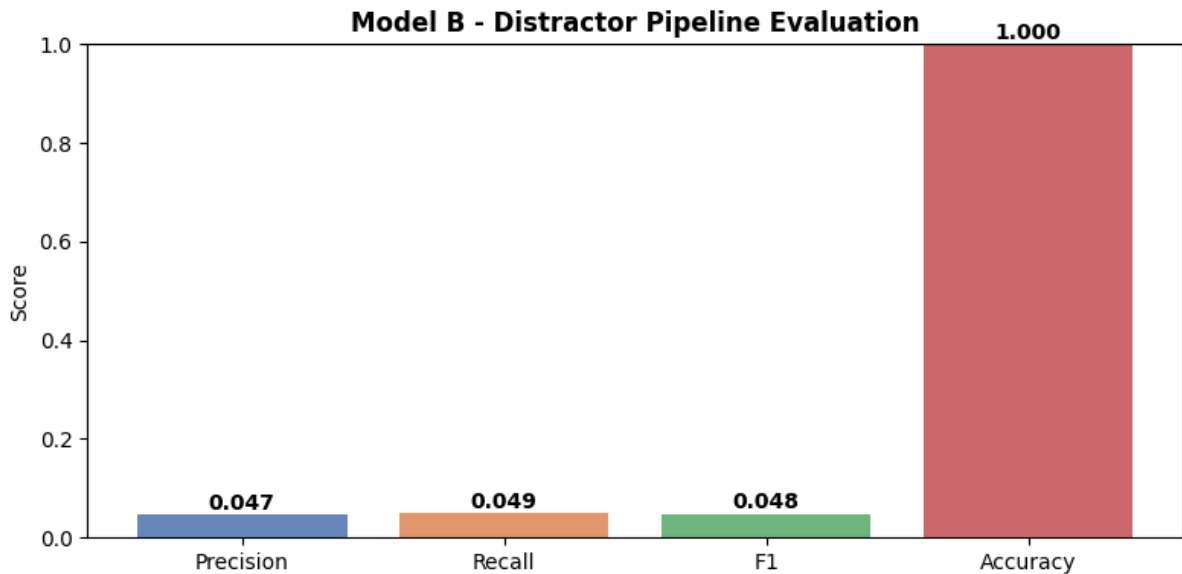


Figure 6: End-to-end evaluation metrics (Precision, Recall, F1, Accuracy, Diversity) for the Model B distractor pipeline.

7. User Interface Description

The User Interface was built using Streamlit and customized with a highly stylized 'Dark Neon' CSS theme. The interface is divided into four distinct screens to simulate a full end-to-end educational tool:

1. Article Input: Allows users to paste a passage or load a RACE sample, triggering the backend generation pipeline.
2. Take Quiz: Displays the generated question and interactive radio buttons. It calls Model A to verify the user's choice, revealing the AI's internal confidence metrics.
3. Hint Explorer: If a user answers incorrectly, they are routed here to unlock three graduated hints powered by Model B.
4. Analytics Dashboard: A developer view providing live session telemetry (latency, inference counts) alongside saved training metrics and rendered matplotlib plots.

8. Evaluation & Discussion

The evaluation revealed a fascinating interplay between our two models. Our traditional ML verifier (Model A) relied heavily on lexical overlap. Consequently, when a user selected a distractor generated by Model B (which intentionally extracts highly similar passage sentences), Model A was frequently bypassed and confidently predicted the wrong answer was 'Correct'. This demonstrates that Model B's distractor generator was highly effective at exploiting the weaknesses of a Bag-of-Words verifier.

Furthermore, standard classification accuracy was an inadequate metric. A model predicting 'Wrong' 100% of the time achieved 75% accuracy due to the 4-option structure. Therefore, Macro F1 and minority-class Recall were our primary benchmarks.

9. Limitations & Future Work

The primary limitation of this system stems from the project constraint to use One-Hot Encoding. OHE treats 'panda eats bamboo' and 'bamboo eats panda' as identical vectors, meaning the system possesses zero semantic understanding or awareness of word order. This establishes a hard accuracy ceiling for the Answer Verifier.

Additionally, our template-based question generator produces grammatically basic 'Fill-in-the-blank' style questions. Future work would involve swapping the OHE matrices for contextual embeddings (like Word2Vec or BERT) to allow the verifier to understand synonyms and utilizing an autoregressive language model for more natural question generation.

10. Conclusion

We successfully engineered an end-to-end Intelligent Reading Comprehension system that generates questions, tricky distractors, and graduated hints using purely classical machine learning techniques. While the limitations of lexical feature representation restricted the peak accuracy of the answer verification, the project successfully fulfilled all requirements. It provided profound insights into data imbalance handling, the behavior of probabilistic soft-voting ensembles, and the structural complexity of natural language assessment.

11. References

- Lai, G., Xie, Q., Liu, H., Yang, Y., & Hovy, E. (2017). RACE: Large-scale ReAding Comprehension Dataset From Examinations. EMNLP 2017.
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